Differential Analysis of LLMs in Text2SQL Generation

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Abstract

Large Language Models (LLMs) have shown remarkable capabilities in translating natural language questions into executable SQL queries, a task known as Text2SQL. In this work, we conduct a systematic evaluation of several state-of-the-art LLMs and two distinct Text2SQL frameworks: agentless and agent-based pipelines. Our experiments assess accuracy, consistency, and execution error on the BIRD-SQL benchmark. By enabling users to query structured databases through natural language, effective Text2SQL systems have the potential to democratize data access, allowing non-developers to retrieve complex information from structured databases without needing specialized technical skills. Our study aims to inform future design choices for building more reliable and accessible natural language interfaces to databases.

1 Introduction

As large language models (LLMs) continue to improve, an interesting application of LLMs is in the generation of SQL queries from natural language. This is known as Text2SQL. Text2SQL can enable non-experts to extract valuable information from a database without having to learn complex query languages like SQL. By simply describing the data they want in plain English, users can leverage LLMs to translate their requests into SQL queries that interact directly with a database. This has the potential to democratize data access, reduce reliance on developers, and improve overall efficiency in data-driven decision-making.

In this project, we compare the performance of different LLMs in generating SQL queries from natural language. We explore 2 different frameworks, agentless and agent-based, and 3 different LLMs, Claude 3.5 Sonnet, GPT-4o mini, and Llama 3.1 8B. In the agentless framework, the descriptions of all tables in the database (relevant and irrelevant) are passed as context to the LLM when generating the SQL query given the natural language query (NLQ). In the agent-based framework, there are 3 agents that handle specific tasks such as determining the number of relevant tables, retrieving the descriptions of the relevant tables, and generating a SQL query given a natural language query. We run our agentless and agent-based frameworks with the 3 different LLMs on the BIRD-SQL benchmark.

By analyzing various LLMs and frameworks, we hope to understand how model choice and pipeline design impact the accuracy, consistency, and execution success of generated SQL queries. Our findings can help guide the development of more reliable Text2SQL systems, making database interaction easier and more intuitive.

2 Previous Work

Early methods of Text2SQL relied on rule-based systems [4] and semantic parsing [7], which required domain-specific engineering. With the rise of deep learning, LLMs began to dominate, enabling better conversion from natural language to SQL queries.

CHASE-SQL [6] and XiYan-SQL [2] are agentic Text2SQL approaches that leverage LLMs. They both implement multi-agent frameworks with three key phases: retrieving relevant schema information, generating candidate queries, and selecting the most appropriate SQL statement, where each phase may have 1 or more specialized agents. CHASE-SQL uses methods such as divide-and-conquer chain-of-thought (CoT) reasoning, query-plan-based CoT reasoning, and online synthetic example generation to enhance candidate SQL query generation. XiYan-SQL uses supervised fine-tuning of multiple generators and in-context learning (ICL) to produce high-quality and diverse SQL candidates. CHASE-SQL and XiYan-SQL rank 2nd and 4th on the BIRD-SQL benchmark.

While these approaches demonstrate the effectiveness of multi-agent Text2SQL pipelines, there is value in comparing different LLMs and framework designs under a common experimental setting. In our work, we test agentless and agent-based pipelines across multiple LLMs in order to better understand how model choice and pipeline structure impact SQL generation.

3 Methodology

In this section, we outline the methodology used for our differential analysis of LLMs in Text2SQL generation.

3.1 Model Selection

For this study, we selected the three Large Language Models (LLMs) Claude 3.5 Sonnet, GPT-4o mini, and Llama 3.1 8B. These models are widely adopted and are diverse in model size and architecture. Particularily, Claude 3.5 Sonnet has 175 billion parameters and both GPT-4o mini and Llama 3.1 8B have 8 billion parameters [1]. Additionally, Llama 3.1 8B is open-source, while GPT-4o Mini and Claude 3.5 Sonnet are closed-source. By comparing these models, we can evaluate how performance varies not only with model size but also between open-source and closed-source LLMs.

Temperature controls the degree of randomness in the text generated by LLMs during inference [5]. The default temperature for Claude 3.5 Sonnet is 1.0, which results in the maximum level of randomness. For GPT-4o mini, the default temperature is 0.7; however, we increase it to 1.0 to enable greater diversity in the SQL queries generated across three runs, with the hope that at least one of the generations is correct. For Llama 3.1 8B, we keep the default lower temperature since smaller models are more likely to hallucinate when exposed to higher levels of randomness.

3.2 Framework Selection

There are many ways that Text2SQL can be implemented. In this study, we compare the performance difference between 2 frameworks, one that performs without AI agents (agentless) and one that utilizes a series of AI agents (agent-based).

1. **Agentless**: In this framework, an LLM is used to generate the response directly. The NLQ and all table descriptions are provided to the **LLM**. The LLM then generates a SQL query. The agentless pipeline can be seen in Figure 1.

2. **Agent-based**: In this framework, there are 3 distinct agents that each play a different role in the generation of the response. Before this pipeline can be used, all table descriptions must be vectorized and stored in **ChromaDB**. There are three agents: the **Relevance Agent**, **Retrieval Agent**, and **Query Agent**. The NLQ and all table descriptions are provided to the **Relevance Agent**, which determines the number of tables that are relevant, denoted as K. Then, the **Retrieval Agent** retrieves the K table descriptions from ChromaDB that are most similar to the NLQ. This process is similar to the retrieval step in a RAG system and allows us to retrieve the relevant table descriptions. Finally, the NLQ and the retrieved relevant table descriptions are provided to the **Query Agent**, which then generates the SQL query. The agent-based pipeline can be seen in Figure 2.

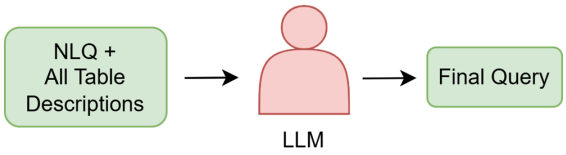


Figure 1: Agentless Text2SQL Pipeline

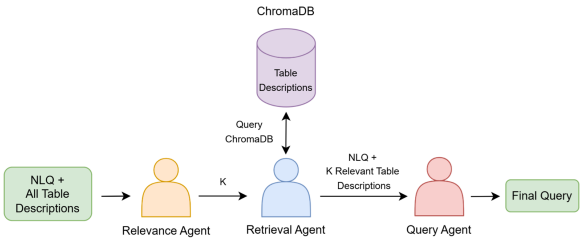


Figure 2: Agent-based Text2SQL Pipeline

The main difference between the agentless and agent-based frameworks is in the SQL generation phase of the pipeline (these are the "red" agents in the architecture diagrams): for agentless, the LLM uses the descriptions of all tables, whereas for agent-based, the query agent uses only the descriptions of relevant tables. This way, we can explore if introducing specialized agents leads to more accurate query generation.

Our agent-based pipeline design is inspired by CHASE-SQL and XiYan-SQL. They divide the Text2SQL pipeline into three main phases: (1) a schema or value retrieval phase to fetch database context, (2) a candidate generation phase to produce potential SQL queries, and (3) a candidate selection phase to choose the best query among multiple options. Our pipeline has the first two phases. We use a Relevance Agent and a Retrieval Agent to retrieve the relevant table descriptions, followed by a Query Agent to generate the final SQL query. However, because we only generate 1 SQL query per NLQ, a selection phase is not necessary since there are no multiple candidates to choose from.

An important motivation for exploring agentless and agent-based frameworks is the "lost in the middle" effect [3], where LLMs struggle to effectively utilize relevant information when it is in the middle of large contexts. As

the context grows—particularly with many irrelevant table descriptions—LLMs can become less accurate in their reasoning if the relevant information gets "lost" in the context. The agent-based framework addresses this by using dedicated agents (Relevance and Retrieval) to retrieve and focus only on the most relevant table descriptions. This reduces the context size and mitigates the "Lost in the middle" effect, with the hope that it can improve the model's ability to generate accurate SQL queries.

3.3 Benchmark Dataset Selection

We use BIRD-SQL as the benchmark for evaluation. It is a widely adopted benchmark in the Text2SQL community, containing cross-domain databases and NLQs of varying complexities (simple, moderate, and challenging). As a result, BIRD-SQL allows for a fair comparison across the different models and frameworks.

The BIRD-SQL includes table descriptions for every database, a collection of NLQs, their corresponding ground-truth SQL queries, and their difficulty levels. For our experiments, we focus on the "student-club" database. This database has 8 tables making the structure complex enough for the problems to be interesting. We sample 30 questions (18 simple, 9 moderate, and 3 challenging) which we evaluate our LLMs and frameworks against.

Note: The BIRD-SQL benchmark has over 1,000 questions, but using all for evaluation is infeasible and costly.

3.4 Evaluation Metrics

We evaluate the performance of each Model+Framework combination across three main metrics: **accuracy**, **consistency**, and **execution error rate**.

- **Accuracy:** Accuracy reflects the correctness of the generated SQL queries. Rather than directly comparing the generated SQL query to the ground-truth SQL query provided by BIRD-SQL (since multiple SQL queries can yield the same result) we determine correctness based on execution results. We execute the generated SQL query and the corresponding ground-truth SQL query from BIRD-SQL on the database, and compare their execution results. A generated query is considered correct if its execution result exactly matches the execution result of the ground-truth query. We categorize correctness based on the number of successful generations out of three attempts for each query: 0/3, 1/3, 2/3, or 3/3 correct generations.

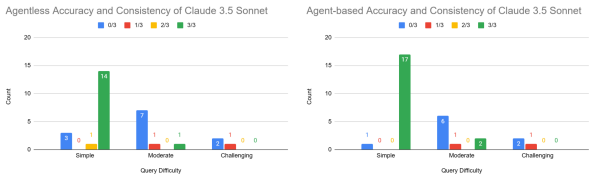
- **Consistency:** We run each NLQ on each Model+Framework combination 3 times. A Model+Framework combination that achieves 3/3 correct generations for a NLQ is highly consistent. On the other hand, fluctuating results (1/3 or 2/3 correct) indicate lower consistency.

- **Execution Error Rate:** This is the percentage of generated SQL queries that fail to execute, typically due to syntax errors, invalid table references, or improper SQL structure. A lower execution error rate reflects syntactic correctness and reliability. A higher execution error rate may indicate hallucinations, misunderstandings of the database schema, or general instability in query generation.

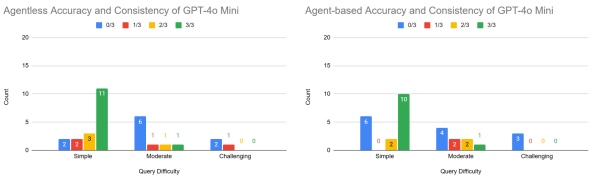
Additionally, we compare the overall performance between the agentless and agent-based frameworks to explore the trade-offs between simplicity (agentless) and modular specialization (agent-based). Each Model+Framework combination is tested on the 30 sampled questions, with three runs per question to capture consistency.

4 Results

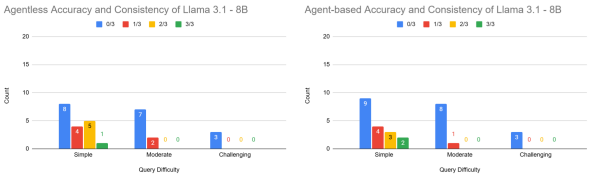
4.1 Accuracy and Consistency



(a) Accuracy and Consistency of Claude 3.5 Sonnet



(b) Accuracy and Consistency of GPT-4o mini



(c) Accuracy and Consistency of Llama 3.1 8B

Figure 3: Accuracy and Consistency

The results from our experiment can be seen in Figure 3. Figure 3a shows the accuracy and consistency results for Claude 3.5 Sonnet. Under the agent-based framework, Claude 3.5 Sonnet achieved near-perfect consistency on simple questions, with 17 having perfect SQL generations (3/3). However, for moderate questions, there was

a significant drop in performance, where 6 had no correct SQL generations (0/3). Performance on challenging questions was worse, with only 1 getting at most 1 correct SQL generation. In the agentless framework, performance was slightly lower than agent-based for simple questions, with 14 having perfect generations. Moderate and challenging questions showed a similar trend of decreased accuracy and consistency. Overall, Claude 3.5 Sonnet demonstrated the highest robustness among all models, especially on simple questions, with the agent-based framework offering a slight advantage.

Figure 3b shows the accuracy and consistency results for GPT-4o Mini. As shown in the graphs, GPT-4o Mini showed more variance even on simple questions. Under the agent-based framework, only 10 simple questions had perfect SQL generations, with 6 having no correct SQL generations. Moderate questions showed mixed success, while challenging questions were unsuccessful. In the agentless framework, GPT-4o Mini's performance on simple questions was slightly better, with 11 having perfect SQL generations. However, the performance on moderate questions was worse. Interestingly, the model was able to generate a correct SQL query for 1 challenging question. Compared to Claude, GPT-4o Mini was less consistent across runs.

Figure 3c shows the accuracy and consistency results for Llama 3.1 8B, which was the least consistent of the three models. Under both frameworks, majority of questions, including simple ones, had no correct SQL generations. Only a small number of questions achieved perfect generations, mainly simple ones. Moderate and challenging questions had the worst results, with most attempts failing across all runs. These results highlight the limitations of smaller or less refined models in SQL generation tasks.

Across all models, simple questions were solved more consistently than moderate and challenging ones. Claude 3.5 Sonnet was the most accurate and consistent model overall, with the agent-based framework performing better. In contrast, GPT-4o Mini and Llama 3.1 8B struggled more. This make sense since Claude 3.5 Sonnet is a significantly larger model. There was no clear difference in performance between agentless and agent-based frameworks for GPT-4o Mini and Llama 3.1 8B. Finally, GPT-4o Mini performs significantly better than Llama 3.1 8B, even though they are approximately the same size. This is probably because factors such as training data quality, fine-tuning strategies, and model architecture, also impact Text2SQL generation.

4.2 Execution Error Comparison

Execution error rate measures the proportion of generated queries that failed to run due to syntax errors, invalid table references, or other structural mistakes. Figure 4 summarizes the execution error rates across models and frameworks.

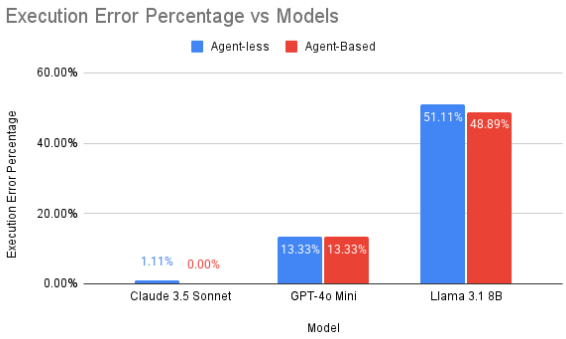


Figure 4: Execution Error Comparison

Claude 3.5 Sonnet had the least error rate of just 1.11% under the agentless framework and 0.00% under agent-based, meaning that it consistently generated syntactically correct SQL queries. GPT-4o Mini had a moderate execution error rate of 13.33% across both frameworks. Llama 3.1 8B had the highest execution error rates, exceeding 48% under both frameworks. This means approximately half of its generated SQL queries had syntax issues or invalid references to columns/tables. This results makes sense since Claude 3.5 Sonnet and GPT-4o Mini are known for their coding abilities.

4.3 Agentless vs Agent-based Performance

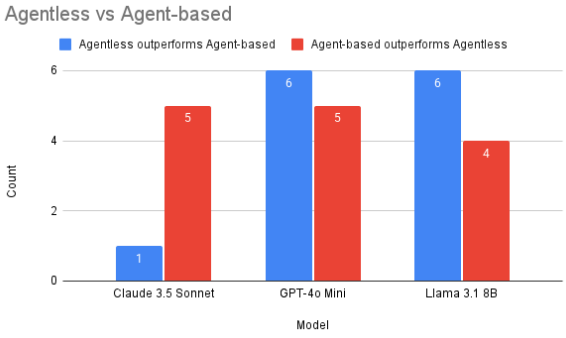


Figure 5: Agentless vs Agent-based Comparison

Finally, we compared the two frameworks by counting how often agentless or agent-based pipelines generated better results for a given query, summarized in Figure 5.

Looking at Claude 3.5 Sonnet, the agent-based framework outperformed agentless in 5 questions, while agentless performed better in only 1 question. This shows that Claude 3.5 Sonnet benefits from our agent-based de-

sign that handles relevant table description extraction. We think this could be because the agent-based system mitigates the "Lost in the Middle" effect. For GPT-4o Mini and Llama 3.1 8B, the agentless framework slightly outperformed agent-based, but there was no significant impact.

Interestingly, Claude 3.5 Sonnet, being a larger model, performed better with the agent-based framework. In contrast, GPT-4o Mini and Llama 3.1 8B, both smaller models, had similar performances between agentless and agent-based pipelines. However, it cannot be definitively concluded that larger models benefit more from agent-based frameworks, because we only tested 1 large model (Claude 3.5 Sonnet). So it is unclear whether the observed improvement is because of Claude's larger size or some other Claude-specific characteristics, such as its training data, fine-tuning methods, or architectural design. Thus, future work would need to evaluate more large models to see if they also show similar performance improvements using the agent-based pipeline.

5 Conclusion

In this study, we conducted a differential analysis of three LLMs (Claude 3.5 Sonnet, GPT-4o Mini, and Llama 3.1 8B) across two different Text2SQL generation frameworks (agentless and agent-based). Our experiments on the BIRD-SQL demonstrated that larger models like Claude 3.5 Sonnet are significantly more accurate, consistent, and robust compared to smaller models. Additionally, the agent-based framework offered slight improvements in performance Claude 3.5 Sonnet, possibly by mitigating issues such as the "Lost in the Middle" effect. However, for GPT-4o Mini and Llama 3.1 8B, which are smaller models, there was no clear advantage between the agent-based and agentless approaches. These findings highlight that both model choice and pipeline structure are important factors in designing Text2SQL systems, especially when targeting real-world applications that require reliability and scalability.

6 Future Directions

In our agent-based framework, we utilized a single Query Agent to generate one SQL query per NLQ. However, leading approaches such as CHASE-SQL and XiYan-SQL employ a multiple generators, producing multiple candidate queries and then selecting the best one. Incorporating multiple Query Agents (by potentially leveraging different LLMs) could help reduce hallucinations and improve overall accuracy.

Our study included only one large-scale model (Claude 3.5 Sonnet) alongside two smaller models (GPT-4o Mini and Llama 3.1 8B). While Claude had better performance with the agent-based framework, it cannot be definitively concluded that larger models inherently benefit more from agent-based designs. Further experiments on other large models would help determine if the observed improvements are driven by model size.

Finally, while the agent-based framework may offer improvements in accuracy for certain models, it also has higher inference time and computational cost due to the additional LLM calls. Future work may include evaluating the trade-offs between performance gains and resource overheads. This is crucial for understanding the practicality of deploying agent-based Text2SQL systems in real-world applications where efficiency and cost are important factors.

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